HDI Predictor Documentation

Problem-Solution Fit Canvas

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Team ID	SB-GENAI-HDI-2025
Project Name	HDI Predictor – Human Development Index Classification System
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Section	Details
1. Customer Segment(s) [CS]	Policy researchers, NGO program officers, development-economics students, and journalists who need quick, indicator-based country assessments
2. Problems / Pains + Frequency [PR]	Manually computing HDI tiers is slow and repeated often — every time a new country, region, or “what-if” scenario needs to be assessed
3. Triggers to Act [TR]	A new report is due, a funding-prioritization meeting is scheduled, or a student is preparing coursework on development indicators
4. Emotions Before / After [EM]	Before: frustrated and time-pressured. After: confident, because the classification is instant and matches the official UNDP methodology
5. Available Solutions [AS]	Manually referencing published UNDP HDI tables (static, not interactive); building an ad-hoc spreadsheet formula (error-prone, not reusable)
6. Customer Limitations [CL]	Limited time, limited statistical tooling, and no dedicated data-science support for a “quick lookup” task
7. Behavior + Intensity [BE]	Frequently cross-checks multiple data sources before finalizing a classification; high intensity around report deadlines
8. Channels of Behavior [CH]	Online — browser-based dashboards, spreadsheets, PDF reports
9. Problem Root Cause [RC]	No lightweight, interactive tool exists that turns the four core HDI indicators directly into a classification without manual computation
10. Your Solution [SL]	A Flask web app backed by a RandomForestClassifier (97.6% test accuracy) that takes life expectancy, schooling years, and GNI per capita and instantly returns the HDI tier and score, packaged in Docker for easy deployment